

Business Management Platform

Business Domain Blueprint

Version 1.0

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This document defines the business domains that make up the Business Management Platform. It establishes a shared business language (Ubiquitous Language), defines ownership boundaries for business capabilities, and identifies how domains interact.

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1. Introduction

Purpose

This document defines the business domains that make up the Business Management Platform. It establishes a shared business language (Ubiquitous Language), defines ownership boundaries for business capabilities, and identifies how domains interact.

The document serves as the foundation for applying Domain-Driven Design (DDD) during implementation. It complements the Software Design Description (SDD) by focusing on business responsibilities rather than technical implementation.

2. Platform Vision

The Business Management Platform enables organizations to manage multiple businesses and branches using a single identity while centralizing sales, inventory, purchasing, customer management, supplier management, reporting, and administration.

The platform is organised into independent business domains that collaborate through well-defined interfaces and events.

3. Domain Map

Business Management Platform is composed of the following independent domains:

- Identity Domain
- Organization Domain
- Business Domain
- Branch Domain
- Membership & Access Domain
- Product Catalog Domain
- Inventory Domain
- Purchasing Domain
- Sales Domain
- Payment Domain
- CRM Domain
- Reporting Domain
- Notification Domain
- Administration Domain
- Configuration Domain

4. Domain Specifications

Identity Domain

Purpose

Responsible for identifying people using the platform.

Owns

- Identity

- Authentication Providers
- Login Sessions
- Passwords

Responsibilities

- Login
- Logout
- Password Reset
- Google Sign-In
- Account Recovery
- Multi-factor Authentication (Future)

Publishes

- IdentityCreated
- IdentityAuthenticated
- PasswordChanged

Consumes

- InvitationAccepted

Organization Domain**Purpose**

Represents the legal or administrative owner of one or more businesses.

Owns

- Organization
- Organization Settings
- Subscription
- Billing

Responsibilities

- Create Organization
- Manage Subscription
- Organization Branding
- Organization Policies

Publishes

- OrganizationCreated
- SubscriptionUpdated

Business Domain**Purpose**

Represents an operational business belonging to an organization.

Owns

- Business Profile
- Business Settings
- Business Status

Responsibilities

- Create Business
- Business Configuration
- Tax Settings

- Receipt Templates
- Business Preferences

Publishes

- BusinessCreated
- BusinessUpdated

Branch Domain***Purpose***

Represents a physical operating location.

Owns

- Branch
- Branch Settings

Responsibilities

- Branch Management
- Branch Configuration
- Operating Hours

Publishes

- BranchCreated
- BranchActivated

Membership & Access Domain***Purpose***

Controls which identities can access which organizations, businesses, and branches.

Owns

- Membership
- Role Assignment
- Permission Assignment
- Invitations

Responsibilities

- Invite Employee
- Accept Invitation
- Assign Roles
- Resolve Permissions

Publishes

- MembershipCreated
- InvitationSent
- InvitationAccepted

Product Catalog Domain***Purpose***

Manages products available for sale.

Owns

- Products
- Categories
- Brands

- Units
- Pricing
- Promotions

Responsibilities

- Product Management
- Pricing
- Promotions
- Product Search

Publishes

- ProductCreated
- PriceChanged
- PromotionActivated

Inventory Domain***Purpose***

Tracks stock across branches.

Owns

- Inventory
- Stock Adjustments
- Stock Transfers

Responsibilities

- Receive Stock
- Adjust Inventory
- Transfer Stock
- Monitor Low Stock

Publishes

- StockReceived
- StockAdjusted
- LowStockDetected

Purchasing Domain***Purpose***

Handles procurement.

Owns

- Purchase Orders
- Goods Receiving

Responsibilities

- Create Purchase Orders
- Receive Deliveries
- Supplier Ordering

Publishes

- PurchaseOrderCreated
- GoodsReceived

Sales Domain

Purpose

Handles all customer sales.

Owns

- Sales
- Sale Items
- Held Sales
- Receipts

Responsibilities

- POS
- Checkout
- Returns
- Receipt Generation

Publishes

- SaleCompleted
- SaleCancelled
- SaleRefunded

Payment Domain***Purpose***

Processes customer payments.

Owns

- Payments
- Payment Transactions

Responsibilities

- Cash Payments
- M-Pesa
- Card Payments
- Refunds

Publishes

- PaymentReceived
- PaymentFailed
- RefundIssued

CRM Domain***Purpose***

Manages business relationships.

Owns

- Customers
- Customer Groups
- Suppliers

Responsibilities

- Customer Management
- Supplier Management
- Customer Search

Publishes

- CustomerRegistered
- SupplierCreated

Reporting Domain***Purpose***

Produces business intelligence.

Owns

- Reports
- Dashboard Metrics

Responsibilities

- Sales Reports
- Inventory Reports
- Customer Reports
- Profit Reports

Note: The Reporting Domain consumes nearly every event published on the platform.

Notification Domain***Purpose***

Communicates with users.

Responsibilities

- Email
- SMS (Future)
- Push Notifications (Future)

Publishes

- EmailSent

Consumes

- PasswordChanged
- InvitationSent
- SaleCompleted

Administration Domain***Purpose***

Manages platform administration.

Owns

- Users
- Roles
- Permissions
- Audit Logs

Responsibilities

- User Administration
- Role Management
- Permission Management

Configuration Domain

Purpose

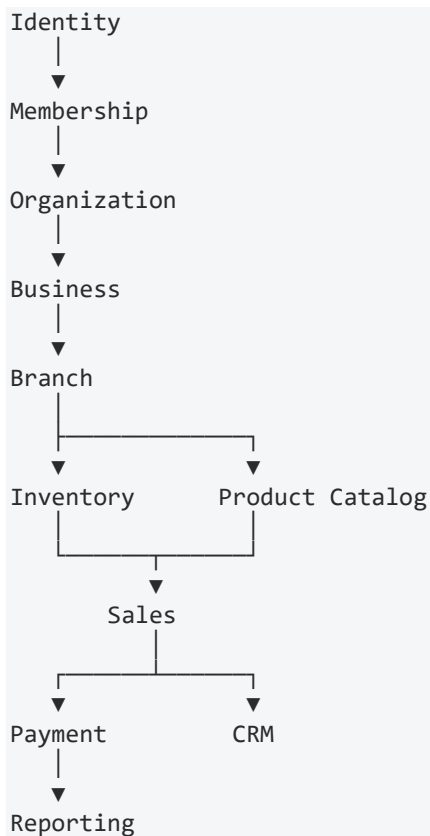
Stores reusable configuration.

Owns

- Receipt Templates
- Number Sequences
- Taxes
- Business Preferences

5. Domain Relationships

The core domains form a dependency chain from identity through to reporting, with Notification observing major events across the platform:



Notification listens to all major events.

6. Ubiquitous Language

Term	Meaning
Identity	A unique person using the platform.
Organization	The legal or administrative owner of businesses.
Business	An operational company managed within an organization.
Branch	A physical operating location of a business.
Membership	A person's relationship to an organization, business, or branch, including their role.

Term	Meaning
Product	An item available for sale.
Inventory	The stock available at a branch.
Purchase Order	A request to procure goods from a supplier.
Sale	A completed customer transaction.
Payment	The settlement of a sale.

7. Domain Events

Event	Published By	Consumed By
IdentityAuthenticated	Identity	Membership
OrganizationCreated	Organization	Business
BusinessCreated	Business	Branch, Configuration
BranchCreated	Branch	Inventory
ProductCreated	Product Catalog	Inventory
PurchaseOrderCreated	Purchasing	Inventory
GoodsReceived	Purchasing	Inventory
SaleCompleted	Sales	Inventory, Payment, Reporting, Notification
PaymentReceived	Payment	Reporting
InvitationAccepted	Membership	Identity
LowStockDetected	Inventory	Notification, Reporting

8. Bounded Contexts (Implementation)

When you start building the backend, each domain maps naturally to a NestJS module:

```
src/
├── identity/
├── organizations/
├── businesses/
├── branches/
├── memberships/
├── catalog/
├── inventory/
├── purchasing/
├── sales/
├── payments/
├── crm/
├── reporting/
├── notifications/
├── administration/
└── configuration/
```

Each module owns its entities, services, controllers, and business rules, while interacting with other modules through well-defined interfaces and events rather than sharing internal logic.